RP (Random Process) $X : \Omega \times T \to \mathbb{R}$ Ω . T can be continous / discrete

Expectation $\mu(t) = E[X(t)]$

Auto Correlation $R(t_1 . t_2) = E[X(t_1) X(t_2)]$ Complex $E[X(t_1) X^*(t_2)]$

Auto Covariance $C(t_1 . t_2) = E[(X(t_1) - \mu_1)(X(t_2) - \mu_2)]$

$$C(t_1 . t_2) = R(t_1 . t_2) - \mu(t_1)\mu(t_2)$$

e.g. $X(t) = A + Bt$ $A . B$ indp , Find C_x , PDF

$$E(X_t) = \mu_A + \mu_B t$$

$$R_x(t_1 . t_2) = E[(A + Bt_1)(A + Bt_2)] = \sigma_A^2 + \sigma_B^2 t_1 t_2$$

$$C_x = R_x - E(X_{t_1}) E(X_{t_2})$$

$$P_x = P_A * \frac{1}{|t|} P_B\left(\frac{x}{t}\right)$$

e.g. $X(t) = T + t$, $T \sim U[0,1]$, Find C_x , PDF

$P_x = 1.$ $x \in [t . t+1]$ $X_t \sim [t . t+1]$

$E(X_t) = \frac{1}{2} + t$ $E(T) = \frac{1}{2}$ $E(T^2) = \frac{1}{4} + \frac{1}{12} = \frac{1}{3}$

$$R_x(t_1, t_2) = E((T + t_1)(T + t_2)) = E(T^2) + (t_1 + t_2) E(T) + t_1 t_2$$

$$= \frac{1}{3} + \frac{1}{2}(t_1 + t_2) + t_1 t_2$$

错误解法 $E(X_{t_1} X_{t_2}) = \iint P_{X_1 X_2} dx dy$

$$C_x(t_1, t_2) = R_x(t_1, t_2) - \left(\frac{1}{2} + t_1\right)\left(\frac{1}{2} + t_2\right) = \frac{1}{12} = D(T)$$

Stationary

SSS (Strict Sense Stationary) : PDF of $X(t)$ time-invariant

WSS (Wide Sense Stationary) : $\mu(t)$ time-invariant , $R(t, t+\tau)$ only about τ

Cyclostationarity $E(X_t) = E(X_{t+kT})$

$$R(X_{t_1}, X_{t_2}) = R(X_{t_1+kT}, X_{t_2+kT})$$

e.g. PAM $X(t) = \Sigma A_k p(t-kT)$

$A_n \in \{-3, -1, 1, 3\}$, indp

$E(X_t) = 0$

$$R(X_{t_1}, X_{t_2}) = E[\Sigma A_i p(t_1 - iT) \cdot \Sigma A_j p(t_2 - jT)]$$

$$= \underbrace{E(A_i^2)}_{\frac{1}{4}\times(9+1+1+9)=5} \Sigma p(t_1 - iT) p(t_2 - iT)$$

e.g. $X \sim N(\mu, \sigma^2)$, $\theta \sim U[-\pi, \pi]$ indp RV

$\cos\alpha\cos\beta = \frac{1}{2}[\cos(\alpha+\beta) + \cos(\alpha-\beta)]$

Prove $Y(t) = X\cos(2\pi f t + \theta)$ WSS

$\sin\alpha\sin\beta = \frac{1}{2}[\cos(\alpha-\beta) - \cos(\alpha+\beta)]$

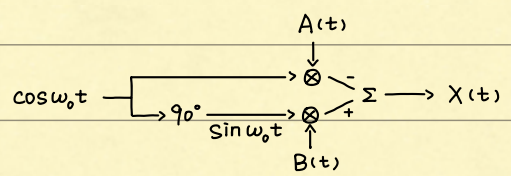
$E(Y(t)) = 0$ since $Y(t) = X(cc - ss)$ $\quad E(\cos\theta) = E(\sin\theta) = 0$

$$E[Y(t_1)Y(t_2)] = \underbrace{E(X^2)}_{\mu^2+\sigma^2} \cdot \underbrace{E(c_1c_2)}_{\frac{1}{2}[\cos(2\pi f(\cancel{t_1+t_2}) + 2\theta) + \cos(2\pi f\tau)]}$$

$$= \frac{1}{2}(\mu^2 + \sigma^2)\cos 2\pi f\tau$$

QAM (Quadrature Amplitude Modulation)

$$X(t) = A(t)\cos\omega t - B(t)\sin\omega t$$



e.g. $A_i, B_i \sim N(0, \sigma^2)$, indp

Prove $X(t) = \sum_{i=1}^{k} A_i\cos\omega_i t + B_i\sin\omega_i t$ WSS

$E(X(t)) = 0$

$R_{\Sigma Y_i} = \Sigma R_{Y_i}$ 算一个即可

$$E(Y_{t_1}Y_{t_2}) = E(A^2)c_1c_2 + E(B^2)s_1s_2 = \sigma^2\cos\omega_i\tau$$

$$R_X(t_1, t_2) = \sigma^2\sum_{i=1}^{k}\cos\omega_i\tau$$